

Retail Futures Trading Strategies Guide

https://gemini.google.com/app/239437be9ea1765d

User prompt: What are some typical forward/future trading strategies for retail investors, e.g. for me using robinhood to speculate a bit?

Response: While forward contracts are private, customized over-the-counter agreements between institutions, retail platforms like Robinhood offer standardized, exchange-traded **futures contracts**. Speculators typically use futures for their nearly 24-hour liquidity, short-selling ease without borrow fees, and built-in leverage.

Common Retail Speculative Strategies

- **Momentum & Trend Following:** Entering in the direction of an established multi-day or intraday move—often using technical indicators like moving average crossovers (e.g., 20/50 EMA) or VWAP. Traders go long in bull trends or short in bear trends across broad indexes (like the S&P 500 or Nasdaq-100) without having to pick individual stocks.
- **Macro Catalyst / Event Trading:** Speculating on short-term price reactions immediately surrounding major economic announcements, such as CPI inflation reports, Federal Reserve rate decisions, or weekly crude inventory numbers. Futures trade overnight and early morning Sunday through Friday, enabling trades while standard stock markets are closed.
- **Key Level Breakouts:** Watching consolidation ranges established during European or overnight sessions. Traders place buy-stop orders above resistance or sell-stop orders below support to capture volatility spikes as US cash markets open.
- **Tactical Portfolio Hedging:** Rather than liquidating long stock positions and triggering capital gains, traders take a short position in an equity index future to offset expected broader market drawdowns over days or weeks.

Contract Sizes: Standard vs. Micro

A standard futures contract controls massive dollar value (notional value). Retail speculators on Robinhood should primarily focus on **Micro contracts**, which trade at 1/10th the size of standard contracts and require substantially lower margin.

Asset Class	Standard Contract	Micro Contract	Micro Point / Tick Value	Approx. Notional Value (Micro)
S&P 500	/ES (\$50/pt)	/MES (\$5/pt)	\$1.25 per 0.25 tick	~\$29,000
Nasdaq-100	/NQ (\$20/pt)	/MNQ (\$2/pt)	\$0.50 per 0.25 tick	~\$40,000
Crude Oil	/CL (1,000 bbl)	/MCL (100 bbl)	\$1.00 per 0.01 tick	~\$7,000
Gold	/GC (100 oz)	/MGC (10 oz)	\$1.00 per 0.10 tick	~\$25,000

Essential Mechanics to Manage

- **Notional Exposure vs. Margin:** Margin in futures is not a loan; it is a performance bond (cash collateral). Putting up \$1,500 in margin to trade one /MES contract means you are effectively managing ~\$29,000 in market exposure. A 3% market drop against your position equals a ~\$870 loss—over half your posted collateral.
- **Hard Stop-Loss Orders:** Because futures trade nearly continuously, sudden overnight headlines can cause steep slippage. Always set stop-losses upon entering an order rather than managing exits manually.
- **Section 1256 Tax Treatment:** In the United States, eligible broad-based index and commodity futures qualify for Section 1256 tax treatment: net capital gains are taxed at 60% long-term and 40% short-term rates, regardless of whether a position was held for six months or six minutes.
- **Auto-Liquidation Thresholds:** If your account balance drops below the broker's maintenance margin requirement, positions are automatically liquidated at market price, frequently locking in the worst possible exit during an intraday dip.

Which underlying asset class—such as equity indices (S&P/Nasdaq), commodities (oil/gold), or Bitcoin—are you considering speculating on first?

Sources

1. [How to Speculate with Futures | Charles Schwab](#)
2. [Futures Trading Strategies: 11 Proven Techniques for 2026](#)
3. [What's a futures contract? - Robinhood](#)
4. [Types of Retail Futures Traders: Day vs. Swing Trading Explained](#)

User prompt: Remind me the basic mechanics? E.g. what are the notion of point/tick values, basic parameters/calculations to manage exposure and understand risk / reward profile etc? Demonstrate with worked out example with details showing upside and downside under different scenarios for a typical strategy

Response: Core Parameters and Mechanics

Understanding a futures contract requires looking past the nominal share price to five foundational mechanics:

- **Contract Multiplier:** The fixed monetary scale assigned to one full integer move (a "point") in the underlying asset. For example, the Micro E-mini S&P 500 (/MES) has a multiplier of \$5 per index point.

- **Tick Size & Tick Value:** A tick is the minimum price increment allowed by the exchange. The tick value is the exact dollar value gained or lost when the price moves one tick:

$$\text{Tick Value} = \text{Tick Size} \times \text{Contract Multiplier}$$

For /MES, the tick size is 0.25 points. Moving one tick changes position value by \$1.25 ($0.25 \times \5). Four ticks equal one point (\$5.00).

- **Notional Value (Total Market Exposure):** The actual dollar amount of the underlying asset you control through one contract:

$$\text{Notional Value} = \text{Index Price} \times \text{Contract Multiplier}$$

If the S&P 500 trades at 5,800, one /MES contract controls \$29,000 ($5,800 \times \5).

- **Initial vs. Maintenance Margin:** Margin is not a loan with borrowing interest; it is a performance bond (good-faith cash collateral).
 - *Initial Margin:* The cash required in your account to open the position (often ~\$1,300 to ~\$1,600 on retail platforms for /MES).
 - *Maintenance Margin:* The minimum balance required to keep the position open. If your account equity falls below this threshold, your broker will automatically liquidate the position at prevailing market rates.
- **Effective Leverage:** The ratio between the total notional exposure and your allocated capital:

$$\text{Leverage} = \frac{\text{Notional Value}}{\text{Account Equity}}$$

Holding a \$29,000 contract with \$2,000 in your account yields **14.5x effective leverage**.

Risk/Reward Management Rules

To protect capital when trading leveraged derivatives, define your risk parameters before entering any order:

1. **Fixed Account Risk:** Risk no more than 1% to 2% of total trading capital on any single setup.
2. **Point Distance to Invalidation:** Determine where your technical trade thesis is proven wrong on the chart, and calculate the distance in points from entry.
3. **Position Size Formula:**

$$\text{Contracts} = \left\lfloor \frac{\text{Maximum Allowed Dollar Risk}}{\text{Stop Distance in Points} \times \text{Contract Multiplier}} \right\rfloor$$

4. **Target Asymmetry:** Require a minimum 2:1 Profit-to-Loss ratio (\$2.00 in projected reward for every \$1.00 risked) to maintain a positive statistical expectancy over time.

Worked-Out Example: Key-Level Breakout Strategy

Account & Setup Parameters

- **Account Size:** \$10,000
- **Risk Cap per Trade (1.5%):** \$150.00
- **Instrument:** 1 contract of /MES (Multiplier: \$5.00/point; 1 tick = 0.25 pts = \$1.25)
- **Initial Margin Required:** \$1,500.00
- **Effective Account Leverage:** 2.9x (\$29,000 notional / \$10,000 account)
- **Technical Context:** The S&P 500 consolidates between 5,785 and 5,800 during the European session. You place a **Buy Stop** order to enter on a breakout above resistance.

Execution Orders

- **Entry Price:** 5,800.00
- **Stop-Loss (Exit on Invalidation):** 5,770.00 (-30.00 points / 120 ticks)

$$\text{Dollar Risk} = 30 \text{ points} \times \$5/\text{point} = \$150.00$$

- **Take-Profit (Target, 2:1 R:R):** 5,860.00 (+60.00 points / 240 ticks)

$$\text{Dollar Reward} = 60 \text{ points} \times \$5/\text{point} = \$300.00$$

Scenario Outcomes & Trade Performance

Parameter	Scenario A: Target Hit (Upside)	Scenario B: Managed Stop (Downside)	Scenario C: Overnight Gap / No Stop (Tail Risk)
Exit Price	5,860.00	5,770.00	5,680.00
Point Move	+60.00 pts	-30.00 pts	-120.00 pts
Gross Dollar P&L	+\$300.00	-\$150.00	-\$600.00
P&L on Margin (\$1,500)	+20.0%	-10.0%	-40.0%
P&L on Total Account (\$10,000)	+3.0%	-1.5%	-6.0%
Operational Result	Trade executed plan cleanly at 2:1 R:R.	Controlled loss within 1.5% risk tolerance.	Incurred heavy slippage or liquidation due to missing hard stop.

In Scenario B, the stop-loss guarantees that a market swing against your trade only costs \$150 (1.5% of capital), even though you controlled \$29,000 of stock index. In Scenario C, holding without a protective stop or across high-impact economic prints allows leverage to work against you, rapidly triggering margin impairment.

What technical setup or timeframe (e.g., 5-minute intraday scalp vs. multi-day swing) are you planning to base your entries and exits on?

User prompt: Typically the points "scale" is the same as the value of the underlying index? E.g. buying S&P micro at entry 5,800 corresponds to the S&P500 itself showing a value of 5,800? And then the notional value is just arrived at by multiplying this value by the multiplier of \$5.00 a point? So effectively in this case we have a 5x leverage? Or how to think about it?

Response: Index futures quote on the exact same numerical scale as the underlying index, but understanding their leverage requires separating the contract's cash multiplier from your account's posted capital.

Points Scale vs. Spot Index

Buying an /MES contract at 5,800 corresponds to the S&P 500 index level around 5,800, though the futures price is rarely an exact 1:1 match with the live cash index (SPX) during the trading day:

- **Cost of Carry (The "Basis"):** The futures price reflects the spot index price plus financing costs (short-term interest rates) minus expected dividend payouts between today and the contract's expiration date.
- **Convergence at Settlement:** Because of continuous institutional arbitrage, the spread between spot and futures stays narrow and steadily decays to zero at expiration.

Notional Value Calculation

The notional value represents the total gross dollar value of the underlying basket of stocks you control:

$$\text{Notional Value} = \text{Futures Price} \times \text{Contract Multiplier}$$

At 5,800 index points with a \$5.00 multiplier:

$$\text{Notional Value} = 5,800 \times \$5.00 = \$29,000.00$$

Every 1.00 point the S&P 500 moves, your position gains or loses exactly \$5.00.

How to Think About Leverage

The \$5.00 multiplier is purely a conversion factor between index points and US dollars—it does **not** mean you have 5x leverage.

Leverage is determined by comparing your total notional market exposure to the cash supporting the trade:

$$\text{Effective Leverage} = \frac{\text{Notional Value}}{\text{Account Capital}}$$

Consider how leverage shifts based on the actual capital backing that single 5,800 /MES contract:

- **Fully Funded (\$29,000 in account):**

$$\frac{\$29,000}{\$29,000} = 1x \text{ leverage (no leverage)}$$

A 1% move in the S&P 500 changes your total account by exactly 1%.

- **Conservative Margin Account (\$10,000 in account):**

$$\frac{\$29,000}{\$10,000} = 2.9x \text{ leverage}$$

A 1% drop in the index (-\$290) equals a 2.9% drawdown on your balance.

- **Minimum Exchange Initial Margin (~\$1,500 posted):**

$$\frac{\$29,000}{\$1,500} \approx 19.3x \text{ leverage}$$

A 1% drop in the index (-\$290) wipes out nearly 20% of your collateral.

The multiplier determines the dollar sensitivity per point, but your leverage is variable and entirely dictated by how much cash you keep in the account relative to the \$29,000 notional footprint.

Given this dynamic, what total cash balance are you planning to hold in the account relative to each micro contract you trade?

User prompt: I have a good amount of cash laying around typically e.g. about \$100k

Response: Holding \$100,000 in cash fundamentally changes your risk profile from high-stakes leverage to fractional exposure. If you trade a single Micro S&P 500 (/MES) contract against a \$100,000 balance, your effective leverage is **0.29x**—meaning your portfolio is roughly one-third as volatile as someone holding \$100,000 of SPY outright.

Sizing and Leverage Scenarios with \$100,000 Cash

At an S&P 500 level of 5,800, here is how different contract quantities scale your actual market exposure and account sensitivity:

Contract Setup	Notional Exposure	Margin Required	Effective Leverage	P&L on 1% S&P Move	Account Impact
1 Micro (/MES)	\$29,000	~\$1,500	0.29x	±\$290	±0.29%
3 Micros (/MES)	\$87,000	~\$4,500	0.87x	±\$870	±0.87%
4 Micros (/MES)	\$116,000	~\$6,000	1.16x	±\$1,160	±1.16%
1 Standard (/ES)	\$290,000	~\$15,000	2.90x	±\$2,900	±2.90%

Key Strategic Considerations for a \$100k Account

- **Immunity to Forced Margin Liquidations:** With 1 /MES, you require ~\$1,500 in margin collateral, leaving over \$98,000 in free equity. An adverse 100-point drop in the S&P 500 reduces your balance by just \$500 (0.5%). You can comfortably place wider, technical stop-losses without worrying about an intraday margin call liquidating your trade.
- **The "Cash-Plus" (Synthetic Equity) Advantage:** In traditional stock trading, gaining \$29,000 of market exposure requires deploying \$29,000 of cash. With futures, you only post ~\$1,500 as a performance bond. The remaining ~\$98,500 can sit in an interest-bearing cash sweep or short-duration Treasury bills, generating 3% to 5% annualized yield while you actively trade.
- **The Oversizing Trap:** The primary psychological danger of having \$100,000 available when trading micros is feeling that single-contract gains are "too small" (\$150–\$300 moves on a \$100k base can feel trivial). This often tempts retail traders to scale up to 10–15 micro contracts or jump into standard /ES before their edge is proven.
- **Account Segregation:** Rather than keeping the entire \$100,000 directly inside an active futures-enabled margin account, consider siloing a smaller dedicated bankroll (e.g., \$5,000 to \$10,000) for your speculative trading. This preserves your cash-plus yield on the bulk of your funds while creating a hard structural circuit breaker against tail-risk software glitches or emotional overtrading.

Are you planning to keep the full \$100k in your brokerage balance to capture interest while trading micros, or would you prefer siloing a smaller, dedicated sub-balance just for speculative execution?

User prompt: Yes planning to keep it in brokerage to capture interest. Question specifically re Robinhood - can you check if they actually offer stop loss order types on futures? I didnt see them in the app

Response: Robinhood does support **stop orders** for futures, but they are easy to miss because of how the mobile interface is structured and how Robinhood handles order tickets.

Where to Find Stop Orders in the App

1. Open the contract page (e.g., /MES) and tap **Trade** → **Sell** (if long) or **Buy** (if short to cover).
2. Look at the **top-right corner** of the order pad where it displays the current order type (it usually defaults to **Market** or **Limit**).
3. Tap that order type text in the top-right corner to open the order types dropdown menu.
4. Select **Stop Order** (or **Stop Limit**).
5. Enter your stop trigger price, the number of contracts, and set the Time in Force (**Day** or **Good 'Til Canceled / GTC**).

Why You Likely Missed It: The Lack of "Bracket" (OCO) Orders

On dedicated futures platforms (like NinjaTrader or Thinkorswim), traders are used to an **"Attach Stop Loss / Take Profit"** toggle directly on the initial buy ticket.

- **No Simultaneous Bracket/OCO:** Robinhood does not support native bracket or One-Cancels-the-Other (OCO) orders for futures. You cannot place an entry order with a built-in stop-loss attached in advance.
- **Two-Step Process:** You must first execute your entry trade (e.g., buy 1 /MES), wait for the fill, and then manually submit a separate resting **Sell Stop** order to define your risk.
- **Single Exit Order Constraint:** Because there is no OCO bracket functionality, you cannot have both an active resting limit order (take-profit) and an active resting stop order (stop-loss) simultaneously covering the *same* contract without locking up additional margin or running the risk of an unintended position flip if both trigger.

CME "Stop with Protection" Behavior

When you submit a stop order on a CME index future like /MES via Robinhood, the exchange executes it as a **Stop with Protection**. Once your trigger price is reached, the exchange turns your stop into a limit order capped at a predefined "protection point" (a set number of ticks beyond the market) rather than an open-ended market order. This protects your account against extreme slippage if the market gaps through your stop during illiquid overnight hours.

If you use **Robinhood Legend** on desktop, stop placement is much more visible: you can right-click the chart to select **Sell Stop** or click the **STP** button directly on the DOM price ladder.

Would you like a walkthrough on how to manage profit targets while keeping a manual stop in place on Robinhood?

Sources

1. [Futures order types: Using market, limit, and stop orders.](#)
2. [Robinhood launched futures trading recently - here's a step ... - Reddit](#)
3. [Does Robinhood Support Bracket Orders? \(And Workarounds\) - Blog](#)
4. [Trading with Robinhood Legend](#)

User prompt: Yes. Give me a realistic example of how to capture some profit in my scenario of having \$100k earning interest in money markets where I want to periodically gain exposure to SP500 directional moves

Response: This framework is known institutionally as a **"Cash-Plus" (or Portable Alpha) strategy**: your \$100,000 cash balance stays parked earning risk-free money market interest, while you use a tiny margin slice (~\$1,500) to opportunistically capture directional S&P 500 swings via 1 Micro contract (/MES).

The Baseline Setup: Yield Engine

- **Cash Allocation:** \$100,000 sitting in your brokerage cash sweep or short-duration Treasury fund earning ~4.5% to 5.0% APY.
- **Passive Monthly Yield:** ~\$375 to ~\$415 per month, completely independent of equity market direction.
- **Available Margin Buffer:** Even with 1 /MES active (~\$1,500 margin requirement), over \$98,500 continues earning yield.

Step-by-Step Trade Execution: The Pullback Swing

Assume the S&P 500 experiences a routine 1.5% pullback to a defined technical support level (such as the 20-day moving average or prior resistance turned support) at **5,800.00**.

1. Entry & Position Sizing

- **Action:** Buy 1 /MES contract at 5,800.00.
- **Total Market Exposure (Notional):** \$29,000 ($5,800 \times \5).
- **Effective Account Leverage:** 0.29x ($\$29,000 / \$100,000$). A 10% market crash would only be a 2.9% hit to your overall account.

2. Managing Risk Without OCO Brackets on Robinhood Because Robinhood does not support simultaneous One-Cancels-the-Other (OCO) orders, you prioritize **capital defense** first:

1. **Submit Protective Stop Immediately:** As soon as your entry fills, place a resting **Sell Stop** order at **5,765.00** (-35 points / -\$175 risk).
2. **Set a Soft Target Alert:** Instead of placing a resting limit sell order (which would tie up additional margin or risk an accidental short if hit alongside your stop), set a price push alert in the app at **5,865.00** (5 points below your target).
3. **Execute the Take-Profit:** When your alert triggers near 5,870.00:
 - Cancel your resting Sell Stop at 5,765.00.
 - Submit a **Limit Sell** at 5,870.00 (or close the position at market).

The Trade Outcome & Settlement

- **Target Exit Price:** 5,870.00 (+70.00 points)

- **Gross Trading Gain:** 70 points × \$5/point = **+\$350.00**
- **Holding Period:** 3 trading days
- **Cash Settlement:** Futures mark-to-market daily. Once closed, the \$350 cash gain settles directly into your brokerage cash balance, expanding your capital base to **\$100,350**.

Monthly Cash-Plus Performance Scenarios

Assuming you execute 3 tactical swing trades per month using this 2:1 reward-to-risk setup (+70 pts / -35 pts):

Monthly Outcome	Active Trading P&L	Cash Sweep Yield (~4.5% APY)	Net Monthly Return	Annualized Impact
Good Month (2 Wins, 1 Loss)	+\$525 (\$700 - \$175)	+\$375	+\$900 (+0.90%)	~10.8% total return
Breakeven Month (1 Win, 2 Losses)	\$0.00 (\$350 - \$350)	+\$375	+\$375 (+0.38%)	~4.5% total return
Tough Month (0 Wins, 3 Losses)	-\$525	+\$375	-\$150 (-0.15%)	~3.0% net loss rate
No Trades (100% Cash)	\$0.00	+\$375	+\$375 (+0.38%)	~4.5% total return

Key Structural Advantages

- **Asymmetric Cushion:** In a poor trading month with three consecutive losses (-\$525), your passive cash sweep interest (+\$375) absorbs nearly 75% of your trading mistakes.
- **Section 1256 Tax Split:** Even if you hold the /MES contract for just two days, that \$350 profit is taxed at **60% long-term / 40% short-term capital gains rates**, significantly lowering your blended tax rate compared to trading individual stocks or non-index ETFs.
- **Low Mental Overhead:** Sizing at 1 micro contract eliminates liquidation anxiety, allowing you to hold through normal intraday chop to reach multi-day swing targets.

What technical catalyst (such as daily moving average pullbacks, RSI extremes, or post-CPI momentum) do you intend to use to trigger your entries?

User prompt: Why do you say it significantly “lowers” blended tax rate compared to trading individual stocks? 60/40% capital gain rates are higher than normal ones in US no?

Response: The "60/40" rule does not mean you pay a 60% or 40% tax rate. Rather, it describes **how your profits are categorized** between long-term and short-term capital gains, regardless of how briefly you hold the trade.

When you trade individual stocks or ETFs and hold them for under a year, the IRS taxes **100% of your profits at ordinary income tax rates** (up to 37% federally). Under Section 1256, futures profits receive a mandatory hybrid split:

- **60% of your profit** is taxed at the lower **Long-Term Capital Gains rate** (typically 15% or 20%).
- **40% of your profit** is taxed at your **Short-Term / Ordinary Income rate** (10% to 37%).

Even if you enter and exit an /MES contract within 15 minutes, 60% of that gain is taxed as if you had held an asset for over a full year.

Blended Tax Rate Comparison by Federal Bracket

Federal Income Bracket	Ordinary Stock / ETF Swing Trade (< 1 yr)	Long-Term Capital Gains Rate	Section 1256 Blended Futures Rate (0.60 × LTCG + 0.40 × Ordinary)	Effective Federal Tax Savings
22%	22.0%	15.0%	17.8%	4.2% lower
24%	24.0%	15.0%	18.6%	5.4% lower
32%	32.0%	15.0%	21.8%	10.2% lower
35%	35.0%	20.0%	26.0%	9.0% lower
37% (Top Bracket)	37.0%	20.0%	26.8%	10.2% lower

(Note: Rates exclude state income taxes and the 3.8% Net Investment Income Tax (NIIT) where applicable.)

Additional Tax Advantages Over Trading Stocks

- **No Wash Sale Rule:** If you take a loss on an individual stock or index ETF (like SPY) and rebuy it within 30 days, the IRS disallows claiming that loss immediately under the wash sale rule. Section 1256 contracts are completely exempt from wash sale restrictions, allowing you to deduct realized losses without waiting 30 days.

- **Simplified Mark-to-Market Reporting:** Instead of tracking dozens of individual buy/sell cost-basis line items on Schedule D at tax time, your broker issues a Form 1099-B with a single aggregate net profit or loss figure for all Section 1256 activity throughout the calendar year.
- **Three-Year Loss Carryback:** If you have a net Section 1256 loss in a bad year, you can elect to carry those losses back up to three prior tax years to offset earlier Section 1256 gains and generate an immediate tax refund, rather than only carrying them forward.

User prompt: What are good screeners to identify 3-4 future contracts worth entering into with this kind of cash plus strategy? E.g. I'm interested in using something like Robinhood or Alpaca brokerage via MCP along with an LLM to automatically drive this strategy. Part of it would be identifying worthwhile contracts and points of entry. What could be a nice way to define this, assuming I'm using an LLM with access to tools/mcp/web search ?

Response: Alpaca does not support futures trading (it is restricted to equities, options, and crypto). While Robinhood supports CME futures in its app and web interface, its public API and MCP capabilities primarily target equities and crypto. For an automated LLM-driven setup, the most reliable architecture pairs an LLM with quantitative market-data MCP tools (such as Databento, Interactive Brokers API, or Tradovate) to screen contracts, using the model to push structured trade alerts or execute directly.

1. The 4-Contract Liquid Micro Basket

To preserve the stability of your \$100,000 cash engine, select micro contracts that offer high liquidity, tight bid-ask spreads, Section 1256 tax treatment, and low cross-asset correlation:

Contract	Asset Class	Multiplier / Tick	Approx. Notional	Role in Cash-Plus Strategy
/MES (Micro S&P 500)	Broad US Equities	\$5.00 / pt (\$1.25/tick)	~\$29,000	Core directional equity exposure; captures institutional market flow.
/MNQ (Micro Nasdaq-100)	Tech / Growth	\$2.00 / pt (\$0.50/tick)	~\$40,000	High beta and momentum; best for clean breakout and momentum swings.
/MGC (Micro Gold)	Precious Metals	\$10.00 / pt (\$1.00/tick)	~\$25,000	Non-correlated hedge against dollar weakness and geopolitical shocks.
/MCL (Micro WTI Crude)	Energy Commodities	\$100.00 / pt (\$1.00/tick)	~\$7,000	Macro supply/demand plays; decoupled from daily equity market movements.

2. Three Quantitative Screeners for Swing Entries

An LLM should not perform raw numerical scanning. Instead, write a lightweight script or MCP tool to scan the 4 contracts against quantitative thresholds, feeding only qualifying setups to the LLM:

A. Trend-Pullback Screener (Mean Reversion in a Trend)

- **Filter:** Daily price > 50 EMA > 200 EMA (Confirms structural bull regime).
- **Trigger Condition:** 4-hour or Daily RSI(14) dips between **38 and 46**, while price tests the 20-day EMA or prior breakout support.
- **Objective:** Buy healthy pullbacks in a prevailing uptrend rather than chasing highs.

B. Volatility Squeeze Screener (Breakout Prep)

- **Filter:** Bollinger Bands contract entirely inside Keltner Channels (John Carter Squeeze indicator), paired with a 14-day Average True Range (ATR) at a 20-session low.
- **Trigger Condition:** Price breaks the prior session's high/low on volume exceeding 1.5× the 20-period volume moving average.
- **Objective:** Capture violent directional expansion following tight consolidations.

C. Macro Divergence & Relative Strength Screener

- **Filter:** Price performance divergence over a rolling 5-day window (e.g., /MES makes a lower low while /MNQ holds a higher low, or /MGC surges while real yields flatten).
- **Trigger Condition:** Positive/negative momentum divergence on the 1-hour MACD histogram at key volume-by-price shelves.
- **Objective:** Identify institutional accumulation before the headline move prints.

3. LLM + MCP Operational Pipeline

Structure your agentic loop in three decoupled layers so that deterministic math handles filtering and the LLM handles context synthesis:

```
[Layer 1: Market Data MCP Tool]
-> Fetches 1h/Daily OHLCV, computes RSI, ATR, EMA, and Squeeze metrics.
-> Filters down to contracts triggering Screener A, B, or C.
    |
    v
```

```
[Layer 2: LLM Reasoning & Macro Context]
-> Calls Web Search MCP to check scheduled high-impact events (CPI, FOMC, EIA Crude).
-> Discards setups where entry occurs < 2 hours before a binary macro release.
-> Calculates precise Stop-Loss, Take-Profit, and 2:1 R:R distance based on ATR.
    |
    v
[Layer 3: Execution / Dispatch MCP]
-> Checks portfolio rule: Total aggregate notional must not exceed $60,000 (0.6x leverage).
-> Submits orders via Broker API OR dispatches an instant Telegram/Discord alert.
```

4. Prompt & Output Schema for the LLM Evaluator

Provide your LLM agent with a strict system instruction to validate candidate contracts returned by the screening tool:

You are a quantitative swing-trading risk manager. You evaluate pre-screened futures setups. Evaluate candidates using these rules:

- Trade Direction must align with the Daily 50 EMA.
- Invalidation point (Stop) must be placed at least 1.5 * ATR away, beyond key structural support/resistance.
- Minimum Profit-to-Loss ratio must be >= 2.0.
- Total account open notional exposure cannot exceed \$60,000 across all trades.

```
Return valid setups in JSON format:
{
  "contract": "/MES",
  "direction": "LONG",
  "entry_price": 5810.00,
  "stop_loss": 5775.00,
  "take_profit": 5880.00,
  "risk_dollars": 175.00,
  "reward_dollars": 350.00,
  "risk_reward_ratio": 2.0,
  "catalyst_notes": "Tested 20-day EMA support with 4h RSI at 41; no major economic releases until PPI on Thu"
}
```

5. System Safety Constraints for Automated Futures Execution

- Exposure Hard Ceiling:** Never allow the agent to exceed 2 active micro contracts simultaneously. Two micro contracts represent roughly ~\$35,000 to ~\$65,000 in notional value, keeping total effective leverage between **0.35x and 0.65x** on your \$100k capital.
- Binary Event Lockout:** Program the tool to automatically block any new entry orders within 30 minutes before and after high-volatility releases (CPI, PPI, FOMC, and Non-Farm Payrolls) to prevent catastrophic slippage.
- Human-in-the-Loop vs. Full Autonomy:** Because of API rate limits and the absence of native bracket/OCO orders on consumer brokers, starting with **semi-automated execution** (where the agent surfaces the setup and calculates the exact stop/target, but you confirm order entry) eliminates execution glitches while preserving the strategy's discipline.

Do you plan to deploy this as an alert-generating copilot where you manually tap the order on your phone, or are you aiming for end-to-end automated execution via a dedicated futures API?

Sources

- [Futures Trading Conditions at Alpaca Trading Explained](#)
- [Agentic Trading on Robinhood | Safe AI Trading Agents](#)

User prompt: Copilot type setup initially. Also a clarifying question - why do we calculate leverage based on overall cash / portfolio size we have? E.g. isnt it more accurate to compare to the margin requirement itself?

Response: Comparing notional value to your margin requirement calculates **contract leverage** (maximum permitted borrowing power), whereas comparing it to your total account equity calculates **effective leverage** (actual economic risk).

Both metrics serve distinct operational functions, but risk managers prioritize effective leverage because it dictates what actually happens to your wealth.

Contract Leverage vs. Effective Leverage

Dimension	Margin-Based Leverage (Contract Leverage)	Equity-Based Leverage (Effective Leverage)
Formula	$\frac{\text{Notional Value}}{\text{Margin Requirement}}$	$\frac{\text{Notional Value}}{\text{Total Available Cash / Equity}}$
Example (\$100k Cash, 1 /MES)	$\frac{\$29,000}{\$1,500} = 19.3x$	$\frac{\$29,000}{\$100,000} = 0.29x$
What It Measures	Minimum collateral constraint and proximity to forced liquidation.	True portfolio volatility and percentage impact on your net worth.

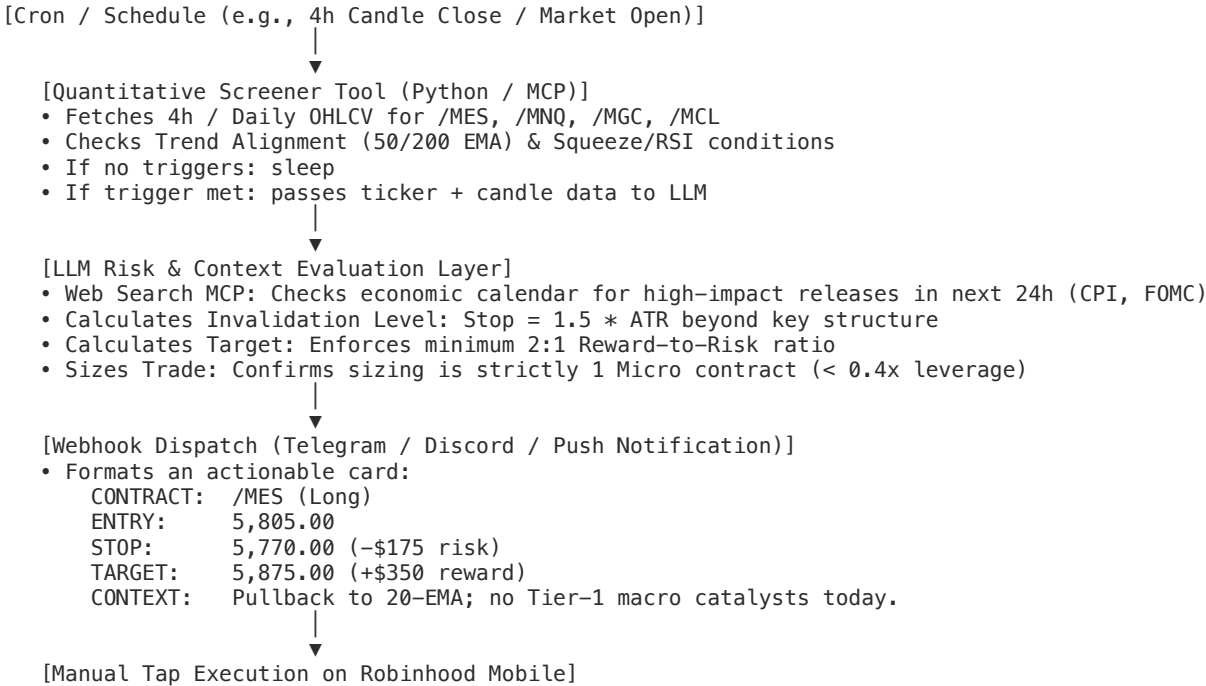
Dimension	Margin-Based Leverage (Contract Leverage)	Equity-Based Leverage (Effective Leverage)
Why It Can Mislead	Margin is an arbitrary performance bond that exchanges change at will without changing your market exposure.	Treats all dollars supporting the risk pool equally, preventing false senses of panic or false safety.

Why Effective Leverage Governs Real Risk

- **P&L Hits Your Net Worth, Not the Margin Bucket:** If the S&P drops 2% (-116 points), 1 /MES loses **\$580**. On a \$1,500 margin requirement, you just lost **38.7%** of your collateral. But you did not lose 38.7% of your capital—your other \$98,500 is completely unaffected, making the actual hit to your portfolio just **0.58%**. Measuring against margin distorts the true severity of the drawdowns.
- **Margin Requirements Are Arbitrary and Variable:** During periods of extreme macro volatility, the CME frequently hikes initial margin requirements overnight—say, from \$1,500 to \$2,500. Your underlying market exposure is still exactly \$29,000, and a 10-point move still equals \$50. If you calculated leverage strictly by margin, your leverage would appear to drop from 19.3x to 11.6x, even though your market risk did not change at all.
- **Preventing "Under-Hedging" and "Over-Positioning":** If you managed positions purely by margin, having \$100,000 might tempt you to think: *"My broker only requires \$1,500 per contract, so I can safely trade 20 contracts."* At 20 /MES contracts, your notional exposure explodes to **\$580,000** (5.8x effective leverage). A routine 3% pullback would wipe out \$17,400—nearly 17.5% of your life savings in hours.

Practical Copilot Architecture for Robinhood Execution

Starting with a copilot model eliminates the risks of API execution failures and order desynchronization. The LLM acts as an analytical filter, delivering pre-calculated, verified setups straight to your phone.

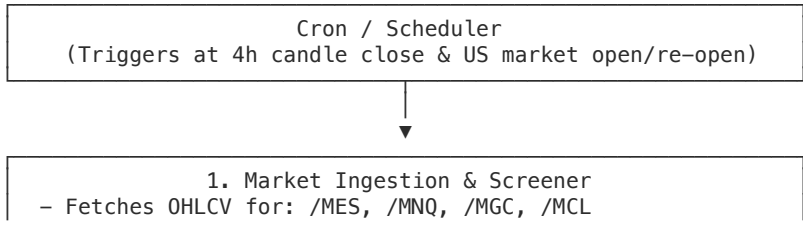


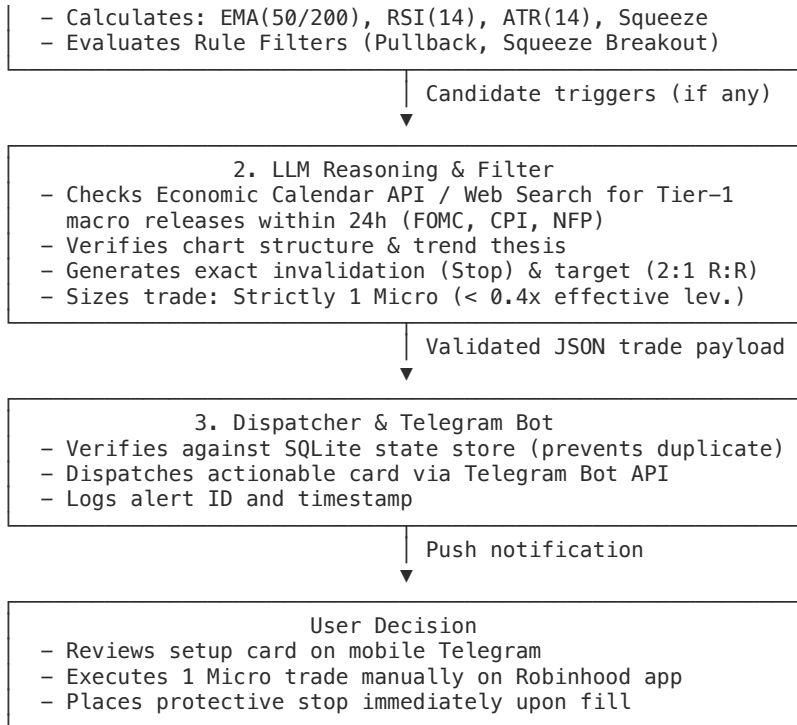
This keeps 100% of the operational control and execution safety in your hands, while delegating chart monitoring, multi-timeframe indicator math, and news checking to the LLM agent.

Which notification channel—such as Telegram, Discord webhook, or local terminal output—would fit most naturally into your daily routine?

User prompt: Likely telegram since they have a pretty good support for programming bots etc. I believe, although slack is also viable. Can you sketch that out for me and then export a full blown design spec encompassing the above ideas into an Md file that I can drop into an agentic coding tool for further development?

Response: System Architecture: Telegram Copilot Flow





Implementation Specification (SPEC.md)

Save the markdown block below as SPEC.md in the root of your project directory for direct consumption by an agentic coding agent.

Cash-Plus Futures Copilot: Technical Specification

1. Executive Overview

An automated personal trading copilot designed for a "Cash-Plus" portfolio (\$100,000 baseline cash generating

2. Core Constraints & Guarantees

1. **Instrument Scope:** Exclusively Micro futures:

- `/MES` (Micro S&P 500)
- `/MNQ` (Micro Nasdaq-100)
- `/MGC` (Micro Gold)
- `/MCL` (Micro WTI Crude)

2. **Fixed Sizing:** Exactly 1 micro contract per signal. Total active open notional exposure across all concu

3. **Reward-to-Risk (R:R):** Strictly ≥ 2.0 . Stop distance must be $\geq 1.5 \times \text{ATR}(14)$ to prev

4. **Macro Event Lockout:** Zero entry alerts permitted within 60 minutes before or 30 minutes after schedule

3. Tech Stack & Dependencies

- **Runtime:** Python 3.11+
- **Task Orchestration:** `apscheduler` or `systemd/cron` daemon
- **Market Data Feed:** Interactive Brokers TWS API, Databento, or Yahoo Finance (`yfinance` for initial stag
- **Technical Analysis:** `pandas`, `pandas-ta` or `numpy`
- **LLM Engine:** Anthropic Claude 3.5 Sonnet / OpenAI GPT-4o via official SDK or LangChain/LiteLLM
- **Telegram Interface:** `python-telegram-bot` (v20+) or direct HTTP calls to Telegram Bot API
- **Persistence:** SQLite3 via `aiosqlite` or `sqlite3` standard library for alert deduplication and signal h

4. System Architecture & Directory Structure

```

cashplus-copilot/ ├── config/ | ├── config.yaml # Thresholds, schedules, asset multipliers | └── .env.example # API keys, Telegram Bot
token, chat IDs ├── src/ | ├── init.py | ├── main.py # Service entry point & scheduler | ├── data/ | | ├── init.py | | └──
market_data.py # OHLCV fetcher & indicator builder | ├── screeners/ | | ├── init.py | | ├── indicators.py # EMA, RSI, ATR,
Keltner/Bollinger formulas | | └── strategies.py # Pullback & Squeeze rule engines | ├── agent/ | | ├── init.py | | └──
calendar.py # Macro/economic event scraper or API | | ├── evaluator.py # LLM prompt orchestration & JSON parsing | | └── prompts.py
# System & task prompts | ├── notifier/ | | ├── init.py | | └── telegram_bot.py # Formatter, message dispatcher, inline callbacks |
  
```

```

└── storage/ | └── init.py | └── db.py # SQLite schemas and deduplication queries |── data/ # Local SQLite DB and logs |──
tests/ | └── test_screeners.py | └── test_evaluator.py |── requirements.txt └── README.md

```

```
---
```

5. Quantitative Screening Algorithms

Strategy A: Trend-Pullback (Mean Reversion)

- **Timeframe:** 4-Hour and Daily candles.
- **Trend Invariant:** Daily Close $> \text{EMA}(50)$ (Long) OR Daily Close $< \text{EMA}(50)$
- **Trigger:**
 - For Long: 4h $\text{RSI}(14)$ crosses below 45 and recovers back above 40 while price is within $\$0.5$ times
 - For Short: 4h $\text{RSI}(14)$ crosses above 55 and falls back below 60 while price is within $\$0.5$ times

Strategy B: Volatility Squeeze Breakout

- **Timeframe:** 1-Hour or 4-Hour candles.
- **Squeeze Condition:** Bollinger Bands (20, 2.0) fully contained inside Keltner Channels (20, 1.5 ATR) for 2
- **Trigger:** First candle closing outside the Bollinger Bands with Volume $> 1.3 \times \text{SMA}(\text{Vol})$

```
---
```

6. Data Schemas

6.1 Screener Candidate Output (Python Dict / Pydantic)

```

python
class ScreenerCandidate(BaseModel):
    contract: str          # e.g., "/MES"
    timeframe: str         # "4h"
    strategy: str          # "TREND_PULLBACK" | "SQUEEZE_BREAKOUT"
    current_price: float   # 5812.50
    ema_50: float          # 5780.25
    ema_200: float         # 5695.00
    rsi_14: float          # 42.30
    atr_14: float          # 28.50
    candle_timestamp: str  # ISO8601

```

6.2 LLM Reasoning & Validation Output (Enforced JSON)

```

{
  "approved": true,
  "rejection_reason": null,
  "contract": "/MES",
  "direction": "LONG",
  "entry_price": 5812.50,
  "stop_loss": 5769.75,
  "take_profit": 5898.00,
  "stop_distance_points": 42.75,
  "target_distance_points": 85.50,
  "risk_reward_ratio": 2.0,
  "risk_dollars": 213.75,
  "reward_dollars": 427.50,
  "effective_leverage": 0.29,
  "macro_clearance": true,
  "thesis_summary": "Daily trend is bullish (Price > 50 > 200 EMA). 4h RSI dipped to 42 and bounced off 20 EMA"
}

```

7. LLM Prompt Architecture

System Prompt

You are the Lead Risk Architect for a conservative futures swing-trading portfolio.
Your capital base is \$100,000 in cash reserves. Capital preservation is your primary directive.

Rules for evaluating trade candidates:

1. Verify trend alignment: Long trades must show price and 50 EMA above the 200 EMA. Short trades must show the opposite.
2. Macro Guardrail: Check the provided economic calendar. If CPI, PPI, FOMC, or NFP are scheduled within 24 hours, reduce position size.
3. Invalidation Calculation: The stop loss must sit at least $1.5 \times \text{ATR}$ away from entry, anchored behind a recent swing point.
4. Profit Target: Must enforce a minimum Reward-to-Risk ratio of 2.0:
 - For Long: $\text{take_profit} \geq \text{entry_price} + (2.0 \times (\text{entry_price} - \text{stop_loss}))$
 - For Short: $\text{take_profit} \leq \text{entry_price} - (2.0 \times (\text{stop_loss} - \text{entry_price}))$
5. Sizing: Always calculate dollar risk for 1 micro contract using these fixed multipliers:
 - /MES: \$5.00 per point

- /MNQ: \$2.00 per point
- /MGC: \$10.00 per point
- /MCL: \$100.00 per point

You must output valid, schema-compliant JSON only. No prose or meta-explanations.

8. Telegram Alert Layout & Formatting

Alerts must be sent using Telegram HTML or MarkdownV2 formatting.

Message Template

 ***TRADE SIGNAL: 1x {contract} ({direction})***
***Strategy:* {strategy}**

 ***Levels***

- ***Entry Price:*` {entry\_price}`**
- ***Stop Loss:*` {stop\_loss}` (-{stop_distance_points} pts | -\${risk_dollars})**
- ***Target (2:1):*` {take\_profit}` (+{target_distance_points} pts | +\${reward_dollars})**

 ***Risk & Portfolio Context***

- ***Capital Risk:* {risk_percent_of_equity}% of \$100k**
- ***Notional Exposure:* ~\${notional_value} ({effective_leverage}x leverage)**
- ***Macro Check:* {macro_status}**

 ***Thesis:***
 {thesis_summary}

 ***Execution Instruction (Robinhood):***

1. Buy/Sell 1 `{contract}` at Market/Limit `{entry\_price}`.
2. Upon fill, immediately submit a resting ***Stop Order*** at `{stop\_loss}` (GTC).

Inline Interactive Buttons

-  **Acknowledge & Tracking** -> Updates local SQLite status to EXECUTED.
-  **Dismiss Signal** -> Updates local SQLite status to DISMISSED.

9. Persistence & Deduplication Schema (SQLite)

```
CREATE TABLE IF NOT EXISTS signals (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
  contract TEXT NOT NULL,
  strategy TEXT NOT NULL,
  direction TEXT NOT NULL,
  entry_price REAL NOT NULL,
  stop_loss REAL NOT NULL,
  take_profit REAL NOT NULL,
  risk_dollars REAL NOT NULL,
  status TEXT DEFAULT 'PENDING', -- 'PENDING', 'EXECUTED', 'DISMISSED', 'EXPIRED'
  telegram_message_id INTEGER
);
```

— Deduplication Rule: Do not emit if a signal for the same contract + strategy exists within 12 hours.

```
CREATE UNIQUE INDEX IF NOT EXISTS idx_recent_signals
ON signals(contract, strategy, timestamp);
```

10. Execution Plan for Coding Agent

1. **Phase 1: Foundation:** Implement `config.yaml`, environment loading, and SQLite database migration in `src/storage/db.py`.
2. **Phase 2: Market Data & Indicators:** Implement `src/data/market_data.py` and `src/screeners/indicators.py` with unit tests for ATR, EMA, RSI, and Squeeze calculations.
3. **Phase 3: Strategy Engine:** Implement candidate generation in `src/screeners/strategies.py`.
4. **Phase 4: Agent & LLM Validation:** Build `src/agent/evaluator.py` with structured JSON output enforcement and calendar checking logic.

5. **Phase 5: Telegram Dispatcher:** Wire `src/notifier/telegram_bot.py` with interactive inline keyboard buttons and formatters.

6. **Phase 6: Orchestration:** Tie the entire pipeline into `src/main.py` using `APScheduler` running every 4 hours aligned with candle closes.

Which market data source do you want to configure first for the quantitative scanner—a free feed like `yfinance`
